

Microscale Phase Change Causes Upscale Energy Transfer in Turbulent Clouds: Evidence from DNS

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Key Points:

- Microscale phase change injects liquid water variance at the small scales, which cascades upscale.
- Turbulence-driven direct cascade and microphysics-driven inverse cascade intersect to produce a spectral break.
- Spatial inhomogeneity and widely separated timescales contribute to the multiscale complexity of atmospheric clouds.

Abstract

The energy spectrum of cloud water content reveals a distinct break, occurring when turbulence-driven direct cascade is disrupted by another cascade. To understand the mechanism of the second cascade, this study applies particle-based direct numerical simulations and models entrainment-mixing in a turbulent cloud. All turbulence scales are explicitly resolved and cloud droplets are tracked in a Lagrangian framework. It is shown that evaporation injects variance into cloud water content at small scales and creates an inverse cascade. Third-order structure functions confirm that turbulence-dominated direct cascade intersects with microphysics-dominated inverse cascade in the liquid water mixing ratio field. Both cascades depend on the intensity of flow turbulence. The gradients of cloud water content follow a trimodal distribution, implying the spatial inhomogeneity of clouds. The timescales governing turbulence and microphysics span ten orders of magnitude. The multiscale behavior of ambient clouds have implications for their parameterization in weather models.

1 Introduction

Atmospheric clouds regulate the Earth's radiative budget and hydrological cycle, yet their internal structure remains one of the difficult features to represent in weather and climate models. The key reason is the enormous range of scales involved. Cloud properties vary coherently from the kilometer scale of convection to the sub-millimeter scale at which individual droplets condense and evaporate. A recurring and informative signature of this multiscale behavior is the appearance of a break in the energy spectrum of cloud water field. This suggests that the power-law exponent changes sharply in the Fourier space and that the regimes on either side of the break are governed by different physical processes.

A closer look at the mechanisms behind the break reveals that microscale cloud growth influences liquid water content (LWC) spectrum. Davis et al. (1999) experimentally measured the LWC in clouds at approximately 4 cm resolution. The corresponding energy spectra showed two distinct power-law regimes, separated by a break. They proposed that an additional variance source exists at small scales but it remained unidentified because of insufficient measurement resolution. In order to theoretically analyze the spectral break, Jeffery (2001) solved an advection-diffusion equation for the LWC with inclusion of phase change. The author found that a production subrange is created in the LWC spectrum by condensation and evaporation. The suggestion that microscale phase change injects variance at small scales is supported by Gotoh et al. (2021). They theoretically showed that the LWC spectrum exhibits two scaling ranges as cloud growth or decay modifies the spectrum. Recently, Wang et al. (2026) observed a physical-space break in the size distribution of growing clouds. Based on large-eddy simulations (LES), they concluded that the break originated from dynamical cloud growth.

A break in the scaling laws can be caused by the intersection of direct and inverse energy cascades. In a laboratory investigation, Xia et al. (2011) showed that energy is transferred from the small to large scales in turbulence if a variance is externally imposed at the small scales. To distinguish between direct and inverse energy cascades, they calculated the third-order velocity structure function. Lilly (1983) suggested that Earth's mesoscale atmosphere has more energy than predicted by theory and the source of extra energy is upscale energy transfer. This hypothesis was tested by Vallis et al. (1997) who performed large-eddy simulations of mesoscale convection and confirmed the presence of inverse energy cascade, although the upscale transfer was reportedly smaller than the downscale transfer. Cabanes et al. (2020) reported that planetary rotation inhibits

vertical motion and creates a stratified atmosphere in which energy is transferred from the small-scale sources up to the large-scale jets.

While the abovementioned studies identified upscale energy transfer and consequent break in the scaling behavior, they did not explore how turbulence-cloud microphysics interactions affect these phenomena. Hence, this study examines the transfer of energy within a turbulent cloud. We apply a particle-based direct numerical simulation (DNS) model which resolves all scales of the flow and tracks cloud droplets individually in Lagrangian fashion. Our model setup is such that it simulates entrainment-mixing between the cloudy and clear air. The grid resolution is much finer than that used in the previous studies of this problem. The objective is to analyze how microscale phase change acts as a localized source of variance and transfers energy from the small to large scales. By contrasting simulations with and without cloud droplets, we isolate the contribution of phase change to upscale energy transfer. We also vary the intensity of flow turbulence and assess its impact on the energy cascades.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Above, $\mathbf{u}$ is the instantaneous air velocity vector, ν is the kinematic viscosity of air, p is the instantaneous air pressure, and ρ_a is the air density. $\mathbf{F}_c$ is an external force that maintains flow turbulence and is constructed in the Fourier space following the procedure of Eswaran and Pope (1988). The buoyancy force $\mathbf{F}_b$ is included but is too weak to sustain turbulence within a small domain. The conservation equations for the temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T + F_T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v + F_{q_v}. \quad (4)$$

Above, L_h is the specific latent heat, C_d is the time rate of exchange between water vapor and liquid water, and c_p is the specific heat of air. μ_T and μ_v are the molecular diffusivities while F_T , and F_{q_v} are the external forcing terms in the temperature and water vapor mixing ratio fields, respectively. The definitions of $\mathbf{F}_c$, $\mathbf{F}_b$, F_T , and F_{q_v} can be found in Sharfuddin et al. (2026). Note that F_T and F_{q_v} are necessary to sustain thermodynamic fluctuations in the respective fields. The environmental supersaturation (S_e) depends on the T and q_v , and is found as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97 \frac{p_{sat}}{p - p_{sat}}} - 1, \quad (5)$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure which can be calculated with respect to water as: (Huang, 2018)

$$p_{sat} = 611.2 \times \exp \left(\frac{17.67(T - 273.15)}{T - 29.65} \right). \quad (6)$$

The equations for the Lagrangian position ($\mathbf{X}$) and velocity ($\mathbf{V}$) vectors of cloud droplets are written as:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad \frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (\tau_p)_i = \frac{2\rho_l(r_i)^2}{9\rho_a\nu}. \quad (7)$$

where the subscript '*i*' denotes the *i*-th particle, *r* is the particle radius, $\mathbf{u}_i$ is the instantaneous air velocity vector interpolated at the position of particle *i*, $\mathbf{g}$ is the gravity vector, τ_p is the particle response time, and ρ_l is the density of liquid water. Cloud droplets grow or shrink according to the following equation:

$$\frac{d(r(t))_i}{dt} = \left(\frac{G}{r(t)} S_e(\mathbf{X}, t) \right)_i, \quad G = \frac{1}{\frac{L_h \rho_l}{k_T T} \left(\frac{L_h}{R_v T} - 1 \right) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}. \quad (8)$$

where *G* is the growth factor, k_T is the thermal conductivity in air, and R_v is the universal gas constant divided by the molecular weight of water. Then, C_d is determined as:

$$C_d(\mathbf{x}, t) = \frac{4\pi\rho_l G}{\rho_a a^3} \sum_{i=1}^n (r(t) S_e(\mathbf{X}, t))_i, \quad (9)$$

where *n* is the number of cloud droplets within a grid cell having a volume of a^3 . The curvature and solute effects are neglected in the present study because cloud droplets remain much larger than dry aerosols throughout the simulation period. Finally, the liquid water mixing ratio q_l is found as:

$$q_l(\mathbf{x}, t) = \frac{4\pi\rho_l}{3\rho_a a^3} \sum_{i=1}^n r_i^3(t). \quad (10)$$

The q_l denotes cloud (liquid) water content and is directly affected by condensational growth or evaporative decay of cloud droplets.

3 Numerical Details

Flow turbulence is assumed to be homogeneous and isotropic. Two turbulent kinetic energy (TKE) dissipation rates are generated by varying the band of forcing wavenumbers. We denote the two TKE levels as 'low' (L) and 'high' (H), respectively. The corresponding wavenumber bands are: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 13.86$ while the Kolmogorov length scales (η) are: 0.0015 m and 0.00088 m. Note that $k_{min} = 2\pi/L = 12.27$ [1/m] is the minimum wavenumber and k_f is the forcing wavenumber. The thermodynamic forcing terms (F_T and F_{qv}) are constructed in the Fourier space with a wavenumber band of $1 \leq k_f/k_{min} \leq 4.12$. More details about external forcing and our particle-based DNS model can be found in Sharfuddin et al. (2026).

The physical model is a cube and has a volume of 0.512^3 m³, with each side having $L = 0.512$ m. The boundary conditions are triply periodic. The domain is equally divided into cloudy and clear-air regions at the start. A slab-like configuration is assumed so that the initial temperature and water vapor mixing ratio change sharply at the cloud/clear-air interface (Gao et al., 2018). The initial vapor mixing ratio is specified as:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases}. \quad (11)$$

Above, $q_v^{max} = 3.949$ g/kg, $q_{v,e} = 0.3$ g/kg, and $d \equiv L/2$ is the width of the cloud slab. q_v^{max} and $q_{v,e}$ represent cloudy (supersaturated) and dry (subsaturated) regions, respectively. The initial temperature field is prescribed as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (12)$$

We simulate three cases: H, L, and H0. The details are provided in Table 1 along with the values of simulated parameters. 16×10^6 cloud droplets are initially placed in the cloudy region, giving a droplet number density of 59.6 cm^{-3} . Entrainment of dry air and turbulent mixing spread cloud droplets to all over the domain. The dominant phase change process is evaporation. The number of grid points is 256^3 and the grid resolution is 2 mm for all cases. We run simulations for a physical time of 20 s. The $k_{max}\eta$ value is 1.38 for Case H and 2.36 for Case L, where $k_{max} = (256\pi)/L = 1570.27 \text{ [1/m]}$ is the maximum wavenumber. The $k_{max}\eta$ measures how accurately the viscous dissipation range is captured in a simulation and is prescribed to be at least greater than 1.0 (Pope, 2000).

Table 1*Details of the test cases*

Case	TKE dissipation rate ϵ_{TKE} (m^2/s^3)	Level of momentum forcing $\mathbf{F}_c$	Initial radius of each cloud droplet (μm)	Stokes number $S_t = \tau_p / \sqrt{(v/\epsilon_{TKE})}$ at initial time	Inclusion of droplets
H	0.0056	High	15	$6.42e - 2$	Yes
L	0.0006	Low	15	$2.12e - 2$	Yes
H0	0.0056	High	15	N/A	No

Case	Description
H	high (H) level of flow turbulence, cloud droplets included
L	low (L) level of flow turbulence, cloud droplets included
H0	high (H) level of flow turbulence, cloud droplets excluded (0)

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_h	2.5×10^6	J/kg	$\mu_T = \mu_v$	2.2×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	c_p	1005.0	J/kg/K
$p(t = 0)$	71446	N/m^2	$\langle T(t = 0) \rangle$	270.75	K
ρ_a	0.92	kg/m^3	ν	1.5×10^{-5}	m^2/s
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K

4 Results and Discussion

This section presents the findings from our particle-based direct numerical simulations. We focus on a single timeframe at 10 s, when velocity and thermodynamic fluctuations are statistically stationary. Because the primary insights of this study are spatial in nature, analyzing one temporal snapshot proves sufficient for our analysis. We investigate the energy spectra, third-order structure functions, and derivatives of key quantities. The timescales associated with the flow and cloud droplets are also examined.

The energy spectra are shown in Figures 1(a)-(f). The E_{TKE} is higher at the large scales for Cases H and H0. This is expected because only the low-wavenumber region is forced. The three E_{TKE} curves converge at the intermediate scales. Difference is visible in the dissipation range between high (H and H0) and low (L) turbulence cases because the $k_{max}\eta$ is smaller for the former. The TKE spectra (Figure 1(a)) are identical for Cases H and H0, suggesting that cloud droplets do not influence flow turbulence. The temperature and water vapor spectra (Figures 1(b)-(c)) do not decay smoothly if cloud droplets are present (Case H and L) but do so otherwise (Case H0). To analyze the reduction of downscale transfer for cases permitting phase change (H and L), we plot the energy spectra of C_d , q_l , and r in Figures 1(d)-(f), respectively. It is evident that microscale evaporation causes upscale energy transfer in the C_d field (Figure 1(d)) and intersection takes place between the direct (downscale) and inverse (upscale) cascades. The intersection marks a break in the scaling behavior since the slope of E_{C_d} is negative (-2.85) to the left of the break and positive (1.2) to the right. Note that a negative slope is indicative of turbulence-dominated direct cascade while a positive slope implies a microphysics-dominated inverse cascade. Similarly, a break is observed in the cloud water spectra (E_{q_l}) (Figure 1(e)). The E_{q_l} peaks in the high-wavenumber region as phase change injects energy at the small scales. If k_b is the wavenumber corresponding to the break, the physical-space break occurs at wavelength $\lambda_b = (2\pi/k_b)$. Using the data of Figure 1(e), $\lambda_b = (2\pi/(10^{-0.25}/0.00088)) \sim 9.8 \text{ mm}$ and $\lambda_b = (2\pi/(10^{-0.2}/0.00153)) \sim 15.23 \text{ mm}$ for Cases H and L, respectively. Interestingly, $\lambda_b/\eta \sim 11.1$ for Case H and $\lambda_b/\eta \sim 9.95$ for Case L. Hence, a break occurs in the cloud water field at a length approximately 10 times the Kolmogorov length. By identifying a break at $\lambda_b \sim 10\eta$, we provide a quantitative validation for Jeffery's (2001) prediction, confirming a production subrange between the inertial and dissipation scales. Since the break is directly related to the Kolmogorov scale, its location depends on the intensity of flow turbulence. We see that the right slope is positive (2.4 and 2.6) for both cases (Figure 1(e)) and corresponds to the high wavenumber region. The left slope, which corresponds to the intermediate wavenumber region, is found to be negative (-0.5) for Case L but positive (0.45) for Case H. A direct cascade (negative slope) exists in the cloud water field for low turbulence but is overwhelmed by the inverse cascade (positive slope) when the turbulence level is high. This is because the direct cascade is weaker and the E_T , E_{qv} , E_{C_d} , and E_{q_l} contain less energy for Case H, indicated by a smaller area under the curve compared to Case L (Figures 1(b)-(e)). Strong flow turbulence enhances mixing and homogenization of the thermodynamic and microphysical fields, reducing energy (variance) in the latter. The radius spectra (E_r) in Figure 1(f) reaffirm the energy injection at high wavenumbers and upscale transfer. Additionally, the E_r exhibits a local peak at $k\eta \sim 10^{-1.5}$ for both cases, implying that direct cascade is present in the low wavenumber region. Since the supersaturation and particle fields are coupled (Equation (8)), large-scale fluctuations in the S_e are transmitted to the particle radius.

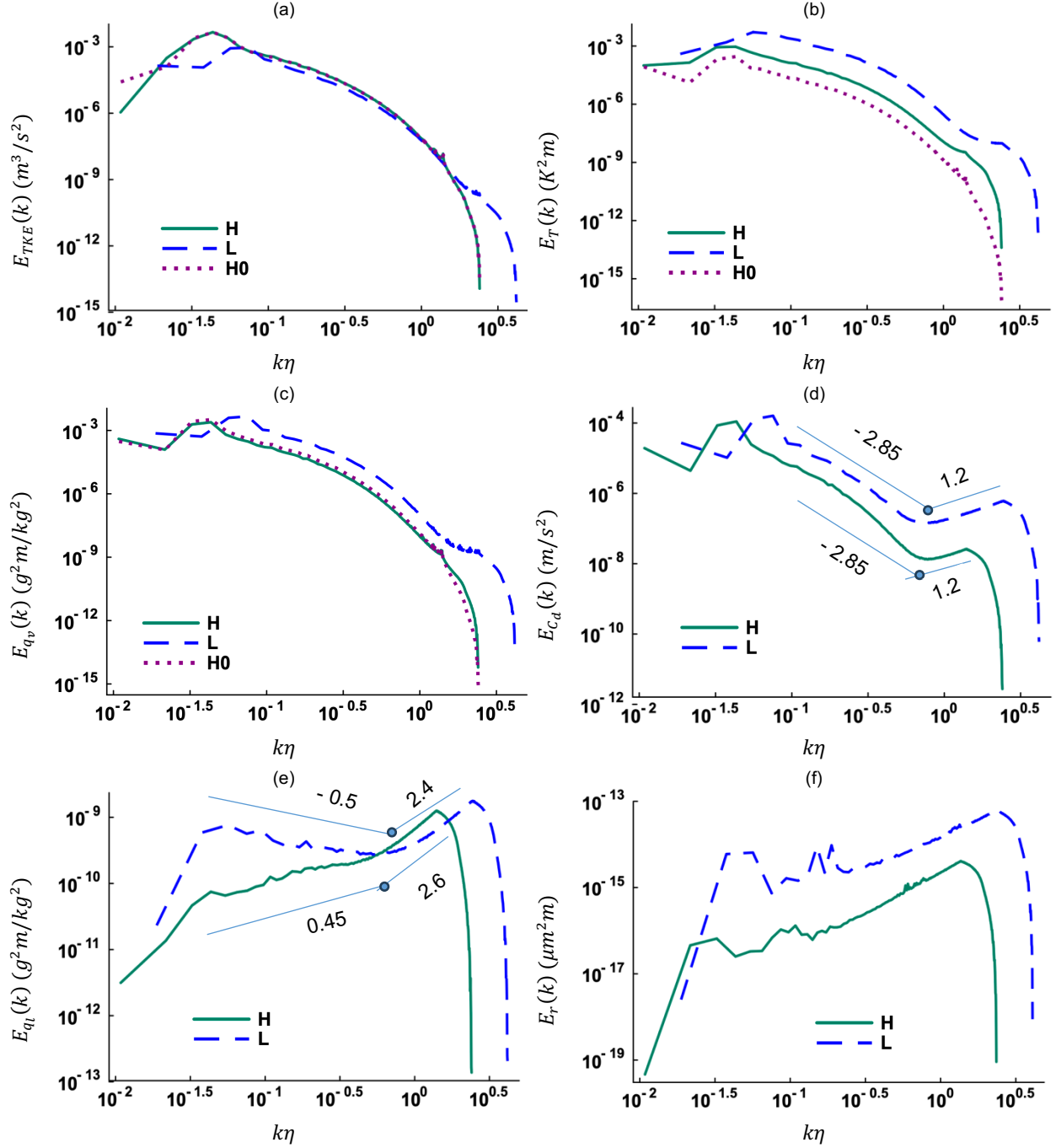


Figure 1. Energy spectra of the (a) turbulent kinetic energy (E_{TKE}), (b) temperature (E_T), (c) water vapor mixing ratio (E_{qv}), (d) phase change term (E_{cd}), (e) liquid water mixing ratio (E_{ql}), and (f) particle radius (E_r). The intersection points in Figures 1(d)-(e) show spectral breaks.

To identify the direction of energy transfer, the third-order structure function (S_3) has been analyzed by Xie et al. (2021), Cho & Lindborg (2001), and Byrne & Zhang (2013). The idea is that a unidirectional energy cascade skews the probability distribution of velocity increments. The third moment preserves the sign of this skewness and indicates whether the cascade is directed upscale or downscale. The third-order structure function for z-velocity (S_3^w) is found as: (Kerr et al., 2001)

$$S_3^w(\Delta z) = - \frac{\langle (w(z + \Delta z) - w(z))^3 \rangle}{\epsilon_{TKE} \Delta z}, \quad (13)$$

where Δz is the separation distance along the vertical direction, w is the z -velocity, and ϵ_{TKE} is the TKE dissipation rate. Note that S_3^w , as formulated in Equation (13), is compensated or unitless. If $S_3^w > 0$, it implies a direct cascade, $S_3^w < 0$ suggests an inverse cascade, and $S_3^w = 0$ means no net energy transfer (Frisch, 1995). Figure 2(a) shows that S_3^w increases with Δz , reaches a peak, and decays, while remaining positive across the entire range of Δz for all three cases. So, energy always flows from the large to small scales in the z -velocity field. Also, the maximum of S_3^w approaches $4/5 = 0.8$, consistent with the exact value derived from Kolmogorov's (1941) $4/5^{\text{th}}$ law. The decay of S_3^w is more rapid when the turbulence level is high (Cases H and H0).

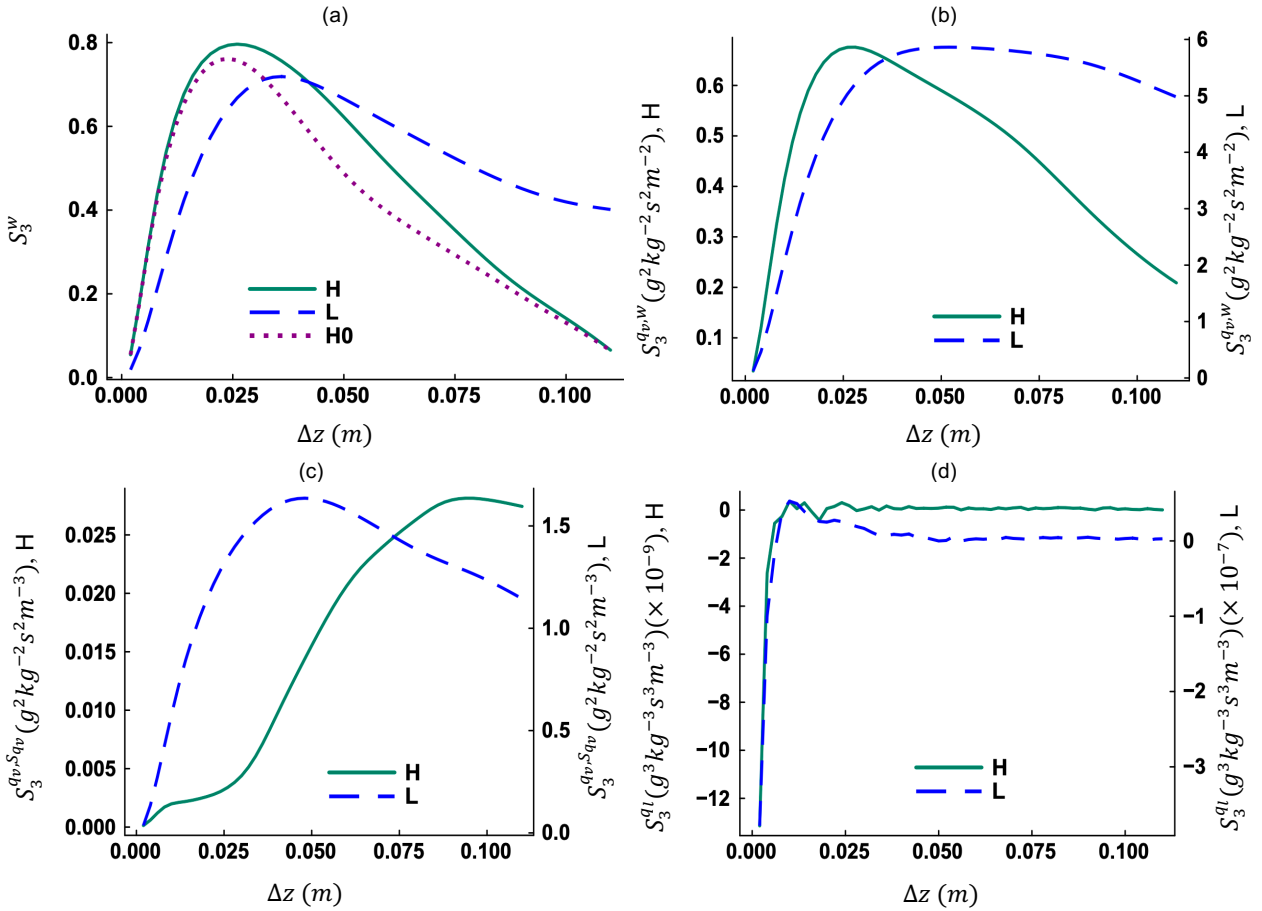


Figure 2. Third-order structure functions (a) S_3^w , (b) $S_3^{qv,w}$, (c) $S_3^{qv,Sqv}$, and (d) S_3^{ql} .

Unlike the velocity field, a passive thermodynamic field does not stretch or fold itself and is simply transported by the velocity field. Hence, the third moment of a pure thermodynamic field, analogous to Equation (13), cannot quantify the advective transfer of thermodynamic variance down to the dissipation scales. Instead, Yaglom (1949) suggested that the S_3 for a thermodynamic field should be calculated as:

$$S_3^{qv,w}(\Delta z) = - \frac{\langle (q_v(z + \Delta z) - q_v(z))^2 (w(z + \Delta z) - w(z)) \rangle}{\epsilon_{TKE} \Delta z}, \quad (14)$$

where the thermodynamic field is water vapor (q_v). Equation (14) contains information about both w and q_v . However, we use ϵ_{TKE} in the denominator instead of the dissipation rate for water vapor ϵ_{q_v} . The microphysical feedback (C_d) modifies the molecular diffusivity for q_v and precludes a precise calculation of ϵ_{q_v} . The external forcing term F_{q_v} along with C_d acts as a source term (S_{q_v}) in Equation (4). To understand the impact of S_{q_v} on energy transfer, we calculate $S_3^{q_v, S_{q_v}}$ as:

$$S_3^{q_v, S_{q_v}}(\Delta z) = - \frac{\langle (q_v(z + \Delta z) - q_v(z))^2 (S_{q_v}(z + \Delta z) - S_{q_v}(z)) \rangle}{\epsilon_{TKE} \Delta z}, \quad S_{q_v} = F_{q_v} - C_d, \quad (15)$$

Figures 2(b) and 2(c) show the $S_3^{q_v, w}$ and $S_3^{q_v, S_{q_v}}$, respectively, and contrast velocity-driven and source-driven transfers in the q_v . Both structure functions are positive and thus inverse cascade is not detected in the q_v . The $S_3^{q_v, w}$ reaches a peak and then decays at a shorter separation distance for Case H compared to Case L. The opposite is true for $S_3^{q_v, S_{q_v}}$. Because turbulent mixing drains water vapor variance toward the dissipative scales, the signature of velocity-driven transfer in high-turbulence conditions (Case H) is confined to shorter spatial separations. The source term counteracts turbulent homogenization by producing variance and does so more actively for Case H, thereby sustaining the source-driven transfer across a broader range of scales.

The q_l is not a transported variable and is directly proportional to r^3 . Variance is injected in the q_l via changes in the r only. Hence, the third order structure function $S_3^{q_l}$ is calculated using the information of q_l only.

$$S_3^{q_l}(\Delta z) = - \frac{\langle (q_l(z + \Delta z) - q_l(z))^3 \rangle}{\epsilon_{TKE} \Delta z}. \quad (16)$$

Figure 2(d) shows that $S_3^{q_l}$ is negative at very small separations for both cases, confirming the presence of inverse cascade or upscale transfer. The $S_3^{q_l}$ does not become positive for Case H as Δz increases, implying the absence of direct cascade and consistent with Figure 1(e). The $S_3^{q_l}$ becomes marginally positive for Case L and then decays to zero with increasing Δz , suggesting a weak direct cascade and again consistent with Figure 1(e). We also determine that $S_3^{q_l}$ reaches zero at $\Delta z \sim 10$ mm and $\Delta z \sim 7$ mm for Cases H and L, respectively.

Independent of phase change, droplet inertia can drive upscale transfer in particle-laden turbulent flows (Brandt & Coletti, 2022). Heavy inertial droplets do not exactly follow the flow and are ejected from high-vorticity regions. They concentrate on low-vorticity zones and such clustering can inject variance in the momentum field that cascades upscale. This is a mechanical pathway of upscale transfer while phase change is a thermodynamic pathway. Since the Stokes number is significantly less than unity in our model (Table 1), the inertia effect is expected to be negligible. To verify this, we calculate the magnitude of flow vorticity ξ as

$$\xi = \left| \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \mathbf{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{k} \right|. \quad (17)$$

The vorticity PDFs are plotted in Figure 3(a). The PDFs are skewed to the right because ξ is always positive. The maximum and mean of ξ are larger if the turbulence level is higher. Most importantly, the PDFs perfectly overlap for Cases H and H0. This confirms that cloud droplet inertia does not affect the local rotation of fluid in our model.

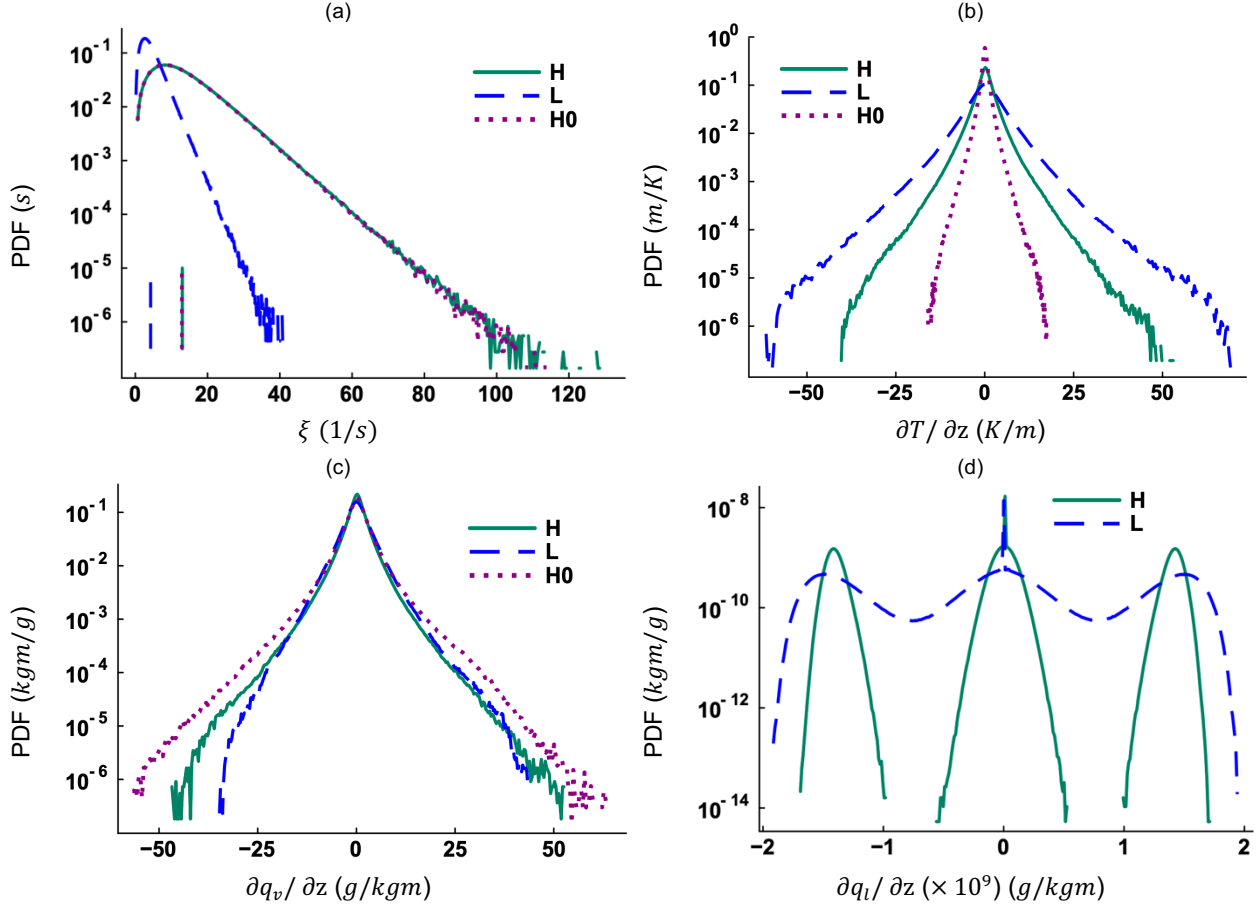


Figure 3. PDFs of the (a) vorticity magnitude, (b) temperature derivative, (c) water vapor mixing ratio derivative, and (d) liquid water mixing ratio derivative. The derivatives are calculated in the z -direction using the central difference formula.

Spatial derivatives quantify the local steepness of a turbulent field. The PDFs of $\partial T / \partial z$, $\partial q_v / \partial z$, and $\partial q_l / \partial z$ are shown in Figures 3(b)-(d), respectively. For cases (H and L) permitting phase change, the distributions of $\partial T / \partial z$ have tails of the stretched-exponential form, which is characteristic of small-scale intermittency. The spread of $\partial q_v / \partial z$ follows the order: $H0 > H > L$ (Figure 3(c)), which is opposite to that for temperature. Evaporation occurs in subsaturated pockets where the q_v is low and adds water vapor to the localized deficits. This smooths out the $\partial q_v / \partial z$. Conversely, evaporation absorbs latent heat from the surrounding environment which is already colder than the warmer subsaturated pocket, thereby steepening the existing temperature gradient. Also, higher turbulence homogenizes thermodynamic fields more efficiently. Thus, intense mixing weakens temperature gradients but enhances water vapor gradients more strongly for Case H compared to Case L. The PDF of $\partial q_l / \partial z$ is found to be trimodal for Cases H and L (Figure 3(d)). The initial domain has equal fractions of cloudy and clear air by volume and subsequent mixing creates pockets of cloudy (supersaturated) and clear air (saturated or subsaturated). The central peak refers to the interior of either cloudy pocket (q_l is approximately uniform) or clear air pocket (q_l is close to zero), where the $\partial q_l / \partial z$ is zero. The left peak (negative $\partial q_l / \partial z$) corresponds to transition from a cloudy pocket outward into clear air, with liquid water content sharply decreasing along the z direction. On the contrary, the right peak (positive $\partial q_l / \partial z$) corresponds to transition

from clear air into a cloudy pocket. It appears that $\partial q_l / \partial z$ varies gradually for Case L but involves discrete jumps for Case H.

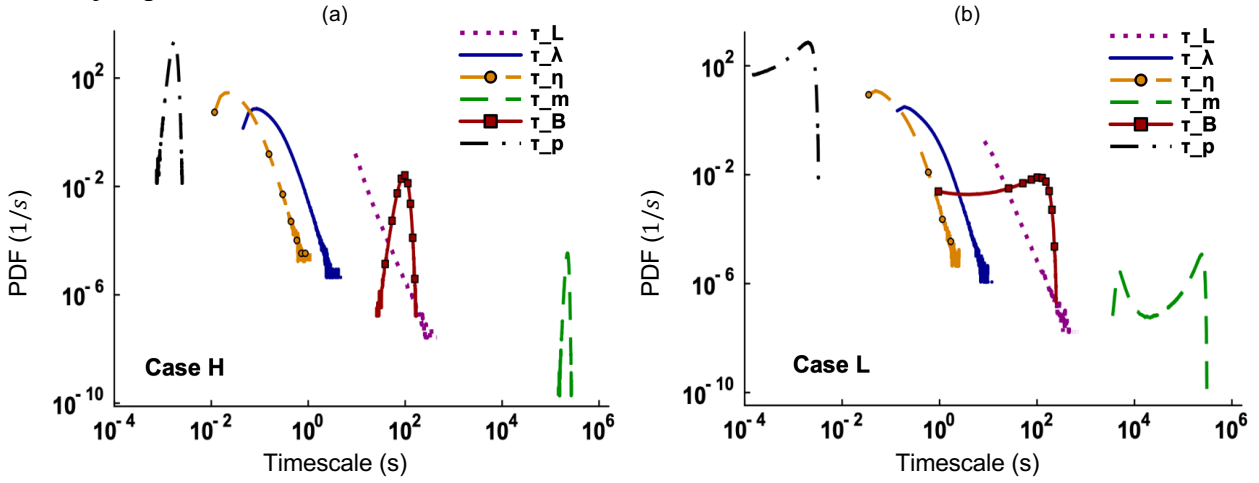


Figure 4. PDFs of the six timescales (τ_L , τ_λ , τ_η , τ_m , τ_B , and τ_p) for (a) Case H and (b) Case L.

It is known that cloud turbulence encompasses a broad range of scales. Hence, we compare the characteristic timescales in our model. The timescale for large eddy turnover is calculated as:

$$\tau_l(\mathbf{x}, t) = \frac{l}{|\mathbf{u}(\mathbf{x}, t)|}. \quad (18)$$

where l is the integral length scale (Pope, 2000). The timescale associated with Taylor microscale is: (Pope, 2000)

$$\tau_\lambda(\mathbf{x}, t) = \sqrt{\frac{15\nu}{\epsilon_{TKE}(\mathbf{x}, t)}}, \quad \epsilon_{TKE}(\mathbf{x}, t) = \nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (19)$$

where ϵ_{TKE} is the TKE dissipation rate. The Kolmogorov timescale is: (Pope, 2000)

$$\tau_\eta(\mathbf{x}, t) = \sqrt{\frac{\nu}{\epsilon_{TKE}(\mathbf{x}, t)}}. \quad (20)$$

The timescale for microscale evaporation is:

$$\tau_m(\mathbf{X}, t) = \frac{(r(t))^2}{|GS_e(\mathbf{X}, t)|}. \quad (21)$$

The timescale for Brownian diffusion is: (Chandrasekhar, 1943)

$$\tau_B(\mathbf{X}, t) = \frac{6\pi\mu(r(t))^3}{k_B T(\mathbf{X}, t)}, \quad k_B = 1.38 \times 10^{-23} \text{ J/K} \quad (22)$$

where k_B is the Boltzmann constant. Additionally, Equation (7) defines the timescale for droplet response.

The six timescales shown in Figures 4(a)-(b) differ by up to ten orders of magnitude and demonstrate the multiscale nature of turbulent clouds. The PDFs of flow timescales (τ_l , τ_λ , τ_η) are skewed to the right in both figures. We see that $\tau_l > \tau_\lambda > \tau_\eta$, which is consistent with $l > \lambda > \eta$.

The droplet response time (τ_p) is much smaller than the flow timescales. The flow field completes

multiple eddy turnovers before a droplet experiences significant evaporation. So, the surrounding environment is uniformly mixed from the perspective of an individual droplet. This condition mathematically corresponds to a Damköhler number ($Da = \tau_\eta / \tau_m$) of less than unity and signifies a homogeneous mixing scenario. Consequently, the τ_m is much larger than the flow timescales, as shown in Figures 4(a)-(b). It also suggests that phase change is quasi-static relative to turbulence and acts as a steady source of variance in the liquid water content. The PDF of τ_m appears to be broader and bimodal for Case L but narrower and unimodal for Case H. High turbulence mixes droplets more effectively with the environment, homogenizing their growth histories and keeping radii closer to the mean value. Low turbulence allows droplets to remain in locally supersaturated or subsaturated conditions for a longer time. Some droplets grow while others evaporate, leading to a broader and bimodal size distribution (Manton, 1979). The timescale for Brownian motion (τ_B) falls between the other two microphysical timescales (τ_p and τ_m). The overlap of τ_B and τ_l is also observed. That is, the time for Brownian motion to randomly displace a droplet by distance r is roughly equal to the time for turbulence to advect droplets over length l . At the macroscale, Brownian diffusion is negligible since $l \gg r$. However, at the microscale, Brownian motion allows droplets to randomly walk across flow streamlines rather than being passive fluid tracers.

5 Implications and Concluding Remarks

The DNS results presented in Section 4 carry several implications for the modeling and observation of cloud microphysics, which are summarized below.

- A) Upscale energy transfer is actively present in atmospheric clouds and is driven by microscale phase change (Figures 1(d)-(f), 2(d)). This inverse cascade originates below the resolution of mesoscale and global models, and remains unresolved in those frameworks. Parameterization of cloud water mixing ratio (q_l) must account for the small-scale production of q_l variance.
- B) The energy spectra, structure functions, and gradients of thermodynamic and microphysical fields are sensitive to the intensity of flow turbulence (Figures 1, 2, and 3). Hence, turbulence parameterization schemes designed for strongly turbulent (cumulonimbus) clouds cannot be directly applied to weakly turbulent (stratocumulus) clouds.
- C) The third-order structure function of q_l is negative at very small spatial separations and crosses the zero line at ~ 7 mm (Figure 2(d)). The grid spacing in a numerical model must be such that this crossover is captured.
- D) The trimodal distribution of $\partial q_l / \partial z$ (Figure 3(d)) suggests that cloudy and clear-air pockets exist within a cloud, even though turbulent mixing is active. Numerical models must account for the spatial inhomogeneity in atmospheric clouds.
- E) The timescales in a cloud span approximately ten orders of magnitude (Figures 4(a)-(b)). Such extreme scale separation makes it infeasible for a single measurement platform to observe the full spectrum of cloud physics.

We conclude that microscale phase change acts as a localized energy source within clouds and drives an upscale energy transfer. Capturing this phenomenon requires a particle-based DNS model which calculates phase change at exact droplet locations and prevents numerical diffusion from artificially destroying local gradients that fuel the inverse cascade. The present study assumes homogeneous and isotropic turbulence. But turbulence in real clouds is often inhomogeneous and anisotropic due to buoyancy and shear. Future iterations of this particle-based DNS model will implement non-periodic boundary conditions to account for buoyancy and shear effects, thereby elucidating their impact on energy cascades in turbulent clouds.

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